

LINUX LAB-1

Que-4

Que 4: Configure file permissions and ownership.

Ans 4: Linux provides `chmod` to change file permissions, `chown` to change file ownership and `chgrp` to change group ownership. File permissions control whether the owner, group and other users can read, write or execute a file. The `ls -l` command is used to verify permissions, owner and group information.

Result

File permissions and ownership were successfully configured and verified using `chmod`, `chown`, `chgrp` and `ls -l`.

```
Ubuntu
x + v
rajat@luffy:/tmp$ mkdir q4dir
rajat@luffy:/tmp$ touch q4dir/file1.txt q4dir/file2.txt
rajat@luffy:/tmp$ ls -l q4dir
total 0
-rw-r--r-- 1 rajat rajat 0 Sep 29 10:48 file1.txt
-rw-r--r-- 1 rajat rajat 0 Sep 29 10:48 file2.txt
rajat@luffy:/tmp$ chmod u+x q4dir/file1.txt
rajat@luffy:/tmp$ ls -l q4dir/file1.txt
-rwxr--r-- 1 rajat rajat 0 Sep 29 10:48 q4dir/file1.txt
rajat@luffy:/tmp$ chmod 640 q4dir/file2.txt
rajat@luffy:/tmp$ ls -l q4dir/file2.txt
-rw-r----- 1 rajat rajat 0 Sep 29 10:48 q4dir/file2.txt
rajat@luffy:/tmp$ sudo chown rajat:rajat q4dir/file1.txt
[sudo: authenticate] Password:
rajat@luffy:/tmp$ ls -l q4dir/file1.txt
-rwxr--r-- 1 rajat rajat 0 Sep 29 10:48 q4dir/file1.txt
rajat@luffy:/tmp$ sudo groupadd q4group
rajat@luffy:/tmp$ sudo chgrp q4group q4dir/file2.txt
rajat@luffy:/tmp$ ls -l q4dir/file2.txt
-rw-r----- 1 rajat q4group 0 Sep 29 10:48 q4dir/file2.txt
rajat@luffy:/tmp$ umask
0022
rajat@luffy:/tmp$ umask 027
rajat@luffy:/tmp$ umask
0027
rajat@luffy:/tmp$ touch q4dir/umaskfile.txt
rajat@luffy:/tmp$ ls -l q4dir/umaskfile.txt
-rw-r--r-- 1 rajat rajat 0 Sep 29 10:53 q4dir/umaskfile.txt
rajat@luffy:/tmp$ sudo chmod -R 750 q4dir
rajat@luffy:/tmp$ sudo chown -R rajat:q4group q4dir
rajat@luffy:/tmp$ ls -lR q4dir
q4dir:
total 0
-rwxr-x--- 1 rajat q4group 0 Sep 29 10:48 file1.txt
-rwxr-x--- 1 rajat q4group 0 Sep 29 10:48 file2.txt
-rwxr-x--- 1 rajat q4group 0 Sep 29 10:53 umaskfile.txt
rajat@luffy:/tmp$ ls -lR q4dir
q4dir:
total 0
-rwxr-x--- 1 rajat q4group 0 Sep 29 10:48 file1.txt
-rwxr-x--- 1 rajat q4group 0 Sep 29 10:48 file2.txt
-rwxr-x--- 1 rajat q4group 0 Sep 29 10:53 umaskfile.txt
rajat@luffy:/tmp$ umask 022
rajat@luffy:/tmp$ umask
0022
rajat@luffy:/tmp$ |
```